Physics
Higher level
Paper 1A and Paper 2
B.4 Thermodynamics

8 November 2026

W4ss Academy practice test 2 | **Topic** B.4 Thermodynamics | **Ref.** W4SS–PHY–B4–02

Student name

30 minutes [Paper 1A and Paper 2]

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- **Paper 1A:** for each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- **Paper 2:** answers must be written within the answer boxes provided.
- A calculator is required for this paper.
- A clean copy of the **physics data booklet** is required for this paper.
- The maximum mark for paper 1A is **[3 marks]**.
- The maximum mark for paper 2 is **[17 marks]**.
- The maximum mark for this examination paper is **[20 marks]**.

Paper 1A

1. Energy is transferred very slowly to ice of mass 0.050 kg at its melting point so that the ice melts completely. The melted water remains at 0 °C.

The specific latent heat of fusion of ice = 335 kJ kg $^{-1}$

What is the entropy change of the ice?

A. 0.041 kJ k $^{-1}$

B. 0.061 kJ k $^{-1}$

C. 0.041 J k $^{-1}$

D. 0.061 J k $^{-1}$

2. A gas undergoes one cycle of a cyclic process.

The net change in internal energy of the gas is

A. zero.

B. positive.

C. negative.

D. determined by the initial temperature of the gas.

3. A thermodynamic process taking place in an isolated system is irreversible when the final state of the system has a:

A. greater number of microstates than the initial state

B. smaller number of microstates than the initial state

C. greater internal energy than the initial state

D. smaller internal energy than the initial state

Paper 1A — answer sheet

Student name _____ Date _____

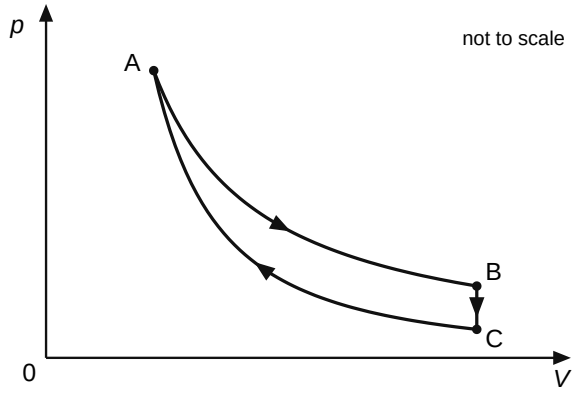
1.	A	B	C	D
2.	A	B	C	D
3.	A	B	C	D

Shade **one** circle per question. If you change your answer, cross out the wrong circle clearly.

Paper 2

Answer **all** questions. Answers must be written within the answer boxes provided.

1. 0.200 mol of an ideal monatomic gas is taken around the cycle A → B → C → A shown on the *pV* diagram. One of the processes between A and B and between C and A is isothermal, the other is adiabatic. Process BC takes place at constant volume.



At A the pressure of the gas is 5.00×10^5 Pa, its volume is 2.00×10^{-3} m^3 and its temperature is 602 K. The pressure at B is 3.00×10^4 Pa and the pressure at C is 4.60×10^3 Pa.

(a) Suggest why AC is the adiabatic part of the cycle. [2]

(b) Show that the volume at C is 3.33×10^{-2} m^3. [2]

(This question continues on the following page)

(Question 1 continued)

(c) Suggest, for the change A ⇒ B, whether the entropy of the gas is increasing, decreasing or constant. [2]

(d) Calculate the thermal energy (heat) taken out of the gas from B to C. [2]

(e) The highest and lowest temperatures of the gas during the cycle are 602 K and 92 K.

The efficiency of this engine is about 0.6. Outline how these data are consistent with the second law of thermodynamics. [2]

2. Ten identical coins are thrown at the same time. Each coin lands heads or tails with equal probability. A macrostate is defined by the number of heads; a microstate specifies the result for each individual coin. The table shows the number of microstates of each macrostate.

Number of heads	0	1	2	3	4	5	6	7	8	9	10
Number of microstates	1	10	45	120	210	252	210	120	45	10	1

(a) Determine the fractional number of throws for which the three most likely macrostates occur. [3]

(b) A throw is made once every minute. Estimate the average time required before a throw occurs where all coins are heads or all coins are tails. [2]

(c) In one throw the coins all land heads upwards. The following throw results in 7 heads and 3 tails. Calculate, in terms of k_B , the change in entropy between the two throws. [2]

Sources of the questions

This practice test was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Physics Questionbank (first assessment 2025), topic B.4 Thermodynamics. The introductions of the Paper 2 questions are missing from the source: W4ss Academy wrote them using only the data of the official markscheme, so that the markscheme still applies. The questions remain the property of the International Baccalaureate Organization and must not be published or sold.

Paper 1A — question 1	EXE.1A.HL.TZ0.22
Paper 1A — question 2	SPM.1A.HL.TZ0.11
Paper 1A — question 3	EXE.1A.HL.TZ0.19
Paper 2 — question 1	23M.2.SL.TZ1.9 — introduction written by W4ss Academy from the markscheme
Paper 2 — question 2	EXE.2.HL.TZ0.14 — introduction written by W4ss Academy from the markscheme

Please **do not** write on this page.

Answers written on this page
will not be marked.